CURRICULUM VITAE

Nathan Hadland

Address:	Arlington, VA (Research Scientist, Arizona Astrobiology Center, The University of Arizona; remote)		
Phone:	480-406-3847	E-mail:	nkhadland@arizona.edu
Website:	https://nkhadland.github.io		

Education:

University of Arizona, Tucson, AZ — May 2026

- Ph.D. Planetary Science, minor Astrobiology, GPA: 4.0 (Defended: December 8th, 2025; Conferred: May 2026)
- Dissertation: "Evaluating the Habitability of Basaltic Volcanic Environments in Iceland as an Analog for Potential Life on Mars" (Advisors: Dr. Solange Duhamel & Dr. Christopher W. Hamilton)

University of Arizona, Tucson, AZ — May 2023

- M.S. Planetary Science, GPA: 4.0

Florida Institute of Technology, Melbourne, FL — May 2020

- B.S. Astrobiology, Summa Cum Laude, GPA: 3.97

Research Summary:

I am interested in microbial ecology across landscapes and scales, and in leveraging those insights to inform astrobiological investigations. My expertise spans environmental microbiology and geobiology across various settings. I have over six years of field experience, including extended campaigns in Iceland, integrating field sampling with remote sensing. In the laboratory, I specialize in microbiology techniques, complemented by computational pipelines in bioinformatics, machine learning, and geospatial data analysis. I am deeply engaged in the astrobiology community, organizing events at The University of Arizona and participating in NASA early-career workshops.

Current Position:

The Arizona Astrobiology Center, Tucson, AZ — January 2026 – Present

Research Scientist

- Plan and lead independent scientific investigations on astrobiology, planetary habitability, biology, and physical sciences. Design analytical and experimental approaches to address novel research questions in space science. Independently develop and submit interdisciplinary technical proposals to federal and private funding sources as principal investigator.
- Direct undergraduate research programs, supervise student and post-baccalaureate researchers, and conduct public outreach/education.

Previous Positions:

The University of Arizona, Tucson, AZ — August 2020 – December 2025

Graduate Research and Teaching Assistantships

- Planned, designed and executed investigations of the habitability and biosignature potential of volcanic systems and the atmosphere (in Iceland, New Mexico, Arizona, Hawaii) as Mars analogs by conducting field work and remote sensing, and adapting experimental and analytical methods to unique low-biomass conditions.

- Developed novel experimental methods for DNA extraction, adenosine triphosphate (ATP) quantification, and bioinformatics workflows and applied Python-based machine learning and statistical-modeling pipelines to interpret complex physical and environmental datasets to inform astrobiology and microbial ecology scientific questions.
- Contributed to multidisciplinary collaborative international field campaigns of up to 30 people and led three-person field deployments; integrated data systems (uncrewed aircraft systems, lidar, differential GPS) to inform scientific investigations in planetary science and astrobiology. Assisted in managing logistics, budgets, permitting, and personnel for multi-week international field campaigns, coordinating across universities and agencies in the U.S.

Florida Institute of Technology, Melbourne, FL Aug 2016 – May 2020
Undergraduate Research Assistant
- Modeled vortex formation and evolution on Neptune

Undergraduate Research Assistant
- Investigated methods to advance *in situ* agriculture on Mars by growing plants in regolith

National Oceanic and Atmospheric Administration, Boulder, CO May 2019 – August 2019
NOAA Hollings Scholarship Program

Lunar and Planetary Institute, Houston, TX June 2018 – August 2018
LPI/NASA Summer Intern Program

University of Tennessee, Knoxville, TN June 2017 – Aug 2017
NSF Research Experience for Undergraduates (REU) Internship

Honors, Awards, and Fellowships:
- 3rd Place in University of Arizona Grad Slam, a university wide graduate student research competition (2026)
- Department of Defense National Defense Science and Engineering Graduate (NDSEG) Fellowship (2022–2025)
- Arizona Space Grant Graduate Fellowship Program (2025–2026)
- Achievement Rewards for College Scientists (ARCS) Foundation Award (2025–2026)
- 2nd Place in Poster Competition at NDSEG Fellowship Conference (2024)
- University of Arizona Galileo Circle Scholarship (2022–2023)
- *Undergraduate:* Astronaut Scholarship Foundation Award (2018 and 2019), NOAA Hollings Undergraduate Award (2018–2020), Florida Institute of Technology Outstanding Junior of the Year Award (*sole recipient selected from the junior class*; 2018)

Publications (in review):
- Calabrese, F., Schroedl, P., **Hadland, N.**, et al., (2026). Compositional Heterogeneity Structures Microbial Microhabitats across Distinct Mineral Substrates. *Nature Communications* (in review).

Publications (see Google Scholar):
- **Hadland, N.**, Hamilton, C.W., Phillips, M., Tuohy, M., and Duhamel, S. (2026). Linking Habitat Complexity to Microbial Distribution in Mars Analog Volcanic Terrains in Iceland. Astrobiology (accepted).

- **Hadland, N.**, Hamilton, C.W., Schroedl, P., Calabrese, F., Marlow, J., and Duhamel, S. (2026). Microbial dispersal from a hyperactive sandsheet in the Icelandic Highland. *Science of the Total Environment 1025, 181659.* https://doi.org/10.1016/j.scitotenv.2026.181659
- **Hadland, N.**, Hamilton, C.W., Björnsdóttir, S.H., and Duhamel, S. (2025). Three eruptions at the Fagradalsfjall volcano in Iceland show rapid and predictable microbial community establishment. *Nature Communications Biology* **8**, 1657, pp. 1–17. https://doi.org/10.1038/s42003-025-09044-1
- Carr, B. Varnam, M., **Hadland, N.**, Shah, J., Voigt, J.R.C., Gwizd, S., Stack, K.M., Calef, F., Francis, R., Basu, U., Björnsson, B., Chen, C.X., Dong, E., Moersch, J.E., Phillips, M., Springer, J., Neish, C.D., and Hamilton, C.W. (2024) Evaluating the use of Unoccupied Aircraft Systems (UAS) for planetary exploration in Mars-analog terrain. *The Planetary Science Journal*, **5**, 231, pp. 1–25. https://doi.org/10.3847/PSJ/ad781e
- Gwizd, S., Stack, K.M., Francis, R., Calef, F., Carr, B.B., Langley, C., Graff, J., Kristinsson, P.H., Thorarensen, V.P., Bernharðsson, E., Phillips, M., Varnam, M., **Hadland**, **N**., Shah, J., Moersch, J., Basu, U., Voigt, J.R.C., and Hamilton, C.W. (2024). Comparing rover and helicopter planetary mission architectures in a Mars analog setting in Iceland. *The Planetary Science Journal*, **5**, 172, pp. 1–35. https://doi.org/10.3847/PSJ/ad55f4
- **Hadland, N.**, Hamilton, C.W. & Duhamel, S. (2024). Young volcanic terrains are windows into early microbial colonization. *Nature Communications Earth and Environment,* **5**, 114, pp. 1–19. https://doi.org/10.1038/s43247-024-01280-3
- Eichler, A., **Hadland, N**., Pickett, D., Masaitis, D., Handy, D., Perez, A., Batcheldor, D., Wheeler, B., and Palmer, A. (2020). Challenging the agricultural viability of Martian regolith simulants. *Icarus*, *354*, 114022, pp. 1–8. https://doi.org/10.1016/j.icarus.2020.114022
- **Hadland, N.**, Sankar, R., Paul LeBeau, R., and Palotai, C. (2020). EPIC simulations of Neptune's dark spots using an active cloud microphysical model. *Monthly Notices of the Royal Astronomical Society, 496, pp.* 4760–4768. https://doi.org/10.1093/mnras/staa1799
- LeBeau, R., Farmer, K., Sankar, R., **Hadland, N**., and Palotai, C. (2020). A numerical investigation of the berg feature on Uranus as a vortex-driven system. *Atmosphere*, *11*(1), pp. 1–11. https://doi.org/10.3390/atmos11010052

Funded Proposals:

- Arizona Astrobiology Center Seed Grant (2024) | $10,000: "Evaluating the Spread of Microorganisms in Terrestrial Sandsheets as a Planetary Analog for Mars"
- Lunar and Planetary Laboratory Director's Grant (2023) | $20,000: "Coriolis Micro Samplers for Planetary Habitability Research"
- Lewis and Clark Fund for Exploration and Field Research in Astrobiology (2022) | $5,000: "Investigating the Habitability of Mars Analog Lava Flow-Fields in Iceland"
- Geological Society of America Graduate Student Grant (2021) | $5,000: "The Origin of Pioneer Microbial Communities Within a Young Lava Flow"
- UArizona Graduate and Professional Student Council (GPSC) Grants (2021–2024) | Total: $9,000: Awarded Research, Travel, and Professional Opportunity grants for fieldwork in Iceland and development of the Astrobiology Journal Club

Pending Proposals as PI:

- "MAP-LT: Morphometric Analysis for Predicting biosignatures in Lava Tubes", NASA Solar System Science (SCUBED), Award Size: Medium (between $150K and $250K average annual budget)
- "(REU Site) Signals of Life: Machine Learning and the Search for Habitability and Biosignatures", NSF Research Experiences for Undergraduates (REU), Award Size: ~$465K

Pending Proposals as Co-I:

- "Ground Truth Space: A Place-Based Analog Reasoning Model for Connecting Local Ecological Inquiry to Planetary Science", NSF K-12, Award Size: ~$400K (PI: Corey Knox, University of Arizona)

Teaching Experience:

- Workshop Designer and Lead, Arizona Astrobiology Center: "Viking 2.0 – Mars Life Detection Workshop," a two-day, team-based data-interpretation workshop for undergraduates, supported by an Arizona Space Grant Fellowship (March 2026)
- Program Lead, Arizona Astrobiology Center Undergraduate Research Cohort, a year-round research and professional development program (2025–Present)
- Teaching Assistant, University of Arizona: Earth, Evolution of a Habitable World (Fall 2020)
- Teaching Assistant, University of Arizona: Exploring our Solar System (Spring 2021, 2022)
- Guest Lecturer, University of Arizona: Life in Extreme Environments (three guest lectures); Mars course (Spring 2026); PTYS 214, Life in the Cosmos (April 2024); and a guest lecture in a course for which I served as teaching assistant

Community Service:

- Subject-Matter Expert, NASA Mars Forward Operational Risk and Contamination Evaluation (M-FORCE) Tiger Team on planetary protection (2026)
- Arizona Astrobiology Center: faculty–student undergraduate research mentor-matching program (2025–Present) and internal seed grant program review (2026)
- Peer reviewer for Scientific Reports, Lewis and Clark Fund for Exploration and Field Research in Astrobiology
- NASA Astrobiology Ideation Factory Member 2023–2025 (Mars Search for Life Working Group)
- Co-Chair of International Geosciences Congress 2024 Planetary Science Sessions
- University of Arizona Astrobiology Journal Club co-lead (2021–Present)
- Arizona Astrobiology Symposium Organizing Committee Member (2024)
- *Undergraduate:* Florida Institute of Technology Astrobiological Research and Education Society – President (May 2019–May 2020), Vice President (May 2018–May 2019)

Post-Baccalaureate Mentorship:

- Noah Fleisher (August 2026–Present): Connecting microbial ecology to remote sensing

Undergraduate Mentorship:

- Parami De Silva (March 2026–August 2026): Astrobiology Atlas: Designing a Spatially Aware Biosignature Database
- Justin Klingele (2022–2023): X-ray computed tomography of pumice

- Samantha Lockwood (2024–2025): 3D internal structure of pumice and its habitability
- Calista Madej (2024–2025): ATP as a biosignature
- Connor Burback (2023–2024): Microorganisms in deserts and perchlorate tolerance

Skills/Certifications:

- *Programming & Machine Learning:* Python (primary), R; machine learning (PyTorch; self-supervised computer vision models and geospatial foundation models, e.g., Prithvi; ensemble methods), bioinformatics pipelines (QIIME2/DADA2), image processing
- *Data & Geospatial Analysis:* Physical and environmental data analysis, QGIS/ArcPro, Cloud Compare and point cloud processing, atmospheric dispersion modeling, Shannon entropy / information-theoretic analysis, drone photogrammetry, LiDAR, differential GPS
- *Other Software:* Dragonfly, Notion, Corel Draw, GitHub, Microsoft products, LaTeX
- *Laboratory, Field, & Instrumentation*: Advanced microbiological laboratory techniques including DNA/RNA extractions, quantitative polymerase chain reactions, adenosine triphosphate analysis, culturing, and flow cytometry. Other analytical techniques include electron microscopy, X-ray diffraction, X-ray computed tomography, aerosol/soil/rock sampling, weather stations, uncrewed aircraft systems. Wilderness First Responder Certified.

Invited Talks:

- Arizona Astrobiology Symposium Plenary Talk (March 27th, 2026): "Bridging Scales and Disciplines: From Fieldwork to Exploration in Astrobiology"
- NASA JPL Planetary Science Seminars (April 7th, 2025): "Rapid Microbial Community Establishment in Newly Emplaced Lava: Implications for the Habitability of Volcanic Terrains on Mars"

Media Engagement and Public Outreach:

- Recurring undergraduate education and career development talks at the Arizona Astrobiology Center (2024–Present)
- Annual volunteer for "Meet MCB" high school student program at the University of Arizona
- NASA OSMA internal news (2026): "M-FORCE Tiger Team Meets to Modernize Planetary Protection" https://sma.nasa.gov/news/articles/newsitem/2026/10/23/m-force-tiger-team-meets-to-modernize-planetary-protection
- NPR 91.5 KJZZ Phoenix (2025): "A volcano in Iceland is helping scientists understand how life forms spread to new habitats" https://www.kjzz.org/science/2025-12-26/a-volcano-in-iceland-is-helping-scientists-understand-how-life-forms-spread-to-new-habitats
- University of Arizona News (2025): "Life on Lava: How Microbes Colonize New Habitats" https://news.arizona.edu/news/life-lava-how-microbes-colonize-new-habitats
- Channel 4 (2024): "National Parks from Above" Documentary Series (Episode 2: Iceland) https://www.channel4.com/programmes/national-parks-from-above
- Astronomy on Tap Space Coast FL (2019), "The Mars Desert Research Station and Growing Plants on Mars"

Field Experience:

- Iceland (2021–Present): Participation in numerous field campaigns, ranging from small teams to groups of over thirty people, including leadership of one field campaign with one additional student. Research activities have focused on astrobiology, active lava

emplacement dynamics, and microbial ecology at Askja volcanic region, 2014–2015 Holuhraun eruption site, and 2021–2023 Fagradalsfjall eruption site.
- Southwest United States (2018–Present): Studying El Malpais National Monument in New Mexico, leading a field campaign with two students to the Grand Falls Dune Field, Arizona, and participating in two missions to the Mars Desert Research Station in Hanksville Utah.
- Hawai‘i (2023): Participation in a field course and collecting UAS data at effusive eruptions from Kīlauea.

Other Publications (Datasets, Code, and White Papers)

- *Dataset:* Carr, B., Hamilton, C. W., **Hadland, N.**, Tuohy, M. (2026) “Hekla (Iceland) Pumice Quarry Unoccupied Aircraft System (UAS) Surveys.” University of Arizona Research Data Repository. https://doi.org/10.25422/azu.data.28915124
- *Dataset:* Hamilton, C.W., Óskarsson, B., Valsson, G., Christoffersen, M.S., Voigt, J.R.C., Moreland, W., **Hadland, N.**, Kubas, A., and Carr, B. (2022) “2021 Fagradalsfjall eruption, Iceland: Ground control points (GCPs).” University of Arizona Research Data Repository. https://doi.org/10.25422/azu.data.21084865
- *Dataset:* **Hadland, N.** and Duhamel, S. (2025) "Rapid microbial colonization of new lava at Fagradalsfjall, Iceland." NCBI Sequence Read Archive, BioProject: PRJNA1205709
- *Dataset:* **Hadland, N.** and Duhamel, S. (2026) "Microbial Dispersal from a Hyperactive Sandsheet in the Icelandic Highlands." NCBI Sequence Read Archive, BioProject: PRJNA1260575
- *Code:* **Hadland, N.** and Phillips. M. (2026) "Habitat Complexity." GitHub repository: https://github.com/nkhadland/Habitat_Complexity_2026
- *Code*: **Hadland, N.** and Duhamel, S. (2025) “ATP Analysis.” GitHub repository: https://github.com/DuhamelLab/ATP
- *White Paper*: Robson, D.J., Grülich, M., Alotaibi, G., Blackburn, H., **Hadland, N.**, and Masaitis, D. (2020): "On the Impact and Needs of Various Audience Groups from Space Analogue Outreach and Education Programmes." University of Leicester Figshare. https://hdl.handle.net/2381/12624101

Conference Abstracts:

- **Hadland, N.**, Hamilton, C.W., Schroedl, P., Calabrese, F., Marlow, J., and Duhamel, S. (2026). “The Atmosphere as a Dynamic Habitat: Microbial Dispersal and Community Structure in Iceland's Hyperactive Dust System.” *Astrobiology Science Conference* (2026)
- Hamilton, C.W., Carr, B.B., Varnam, M., **Hadland, N.**, Shah, J., Voigt, J., Gwizd, S., et al. "Mars Analog Field Testing Reveals Key Operational Strategies for Planetary Exploration Using Unoccupied Aircraft Systems (UAS)." *AGU Fall Meeting* (2025)
- **Hadland, N.**, Hamilton, C.W., Björnsdóttir, S.H., and Duhamel, S. “Development of Early Microbial Communities within Newly Emplaced Lava in Iceland: A 3-Year Fixed-site Time-series Study.” *37th International Geological Congress* (2025)
- **Hadland, N.**, Hamilton, C.W., and Duhamel, S. "Evaluating the habitability of basaltic volcanic environments in Iceland as an analog for potential life on Mars." *Arizona Astrobiology Symposium* (2024)
- **Hadland, N.**, Hamilton, C.W., Björnsdóttir, S.H., and Duhamel, S. “Development of a Microbial Community in a Freshly Emplaced Lava Flow at Fagradalsfjall, Iceland.” *Astrobiology Science Conference* (2024)

- **Hadland, N.**, Hamilton, C.W., and Duhamel, S. “Evaluating the survival and spread of microbes in deserts amid soil-atmosphere interactions.” *NDSEG Fellowship Conference* (2024).
- Shah, J., Carr, B.B., **Hadland, N.**, Varnam, M., Voigt, J.R.C., Basu, U., Björnsson, B., Chen, C., Dong, E., Graff, J., Hibbard, S.M., Moersch, J.E., Phillips, M., Springer, J., Neish, C.D., and Hamilton, C.W. "Evaluating the Use of Unoccupied Aircraft Systems (UAS) for Planetary Surface Exploration in Analog Terrain." *54th Lunar and Planetary Science Conference* (2023)
- Basu, U., Moersch, J., Hamilton, C.W., Scheidt, S., Voigt, J.R.C., Stack, K.M., Francis, R., Calef, F., Golombek, M., and **Hadland, N.** "Selecting Suitable Test Sites at Holuhraun, Iceland, for Mars Mission Simulations Using Rovers and Unmanned Aircraft Systems (UAS)." *53rd Lunar and Planetary Science Conference* (2022)
- Bardabelias, N., Holt, J., Hamilton, C.W., Christoffersen, M.S., and **Hadland, N.** "Characterizing Morphology of Lava Tubes in El Malpais National Monument Using Ground Penetrating Radar and LIDAR." *AGU Fall Meeting* (2021)
- **Hadland, N.**, Sankar, R., Paul LeBeau, R., and Palotai, C. “EPIC Modeling of Large-Scale Dynamical Features of the Gas Giants.” *European Planetary Science Congress - American Astronomical Association Division of Planetary Science Joint Conference* (2019)
- de Boer, G., Cox, C., Lawrence, D., Osborn, J., Hamilton, J., Intrieri, J.M., Maahn, M., Cahill, C.F., Foscue, G., **Hadland, N.**, and Cassano, J.J. "Observatory- and UAS-based perspectives on the surface energy budget of the Arctic." *AGU Fall Meeting* (2019)
- Grulich, M., Alotaibi, G., Blackburn, H., **Hadland, N.**, Larrea Brito, N., Masaitis, D., Robson, D., Triviño, V., and Foing, B. "Goals and first results from MDRS IESL Crew 205." *EGU General Assembly Conference* (2019)
- Le Beau, R.P., Loiacono, R., Wagner, A., Farmer, K., Palotai, C.J., Sankar, R., **Hadland, N.**, and Nodolski, N. "Using Numerical Simulation to Consider the Observed Variations in the Behavior and Appearance of Dark Spots on the Ice Giants." *American Astronomical Society Meeting* (2019)
- LeBeau, Jr., R.P., Farmer, K., Palotai, C.J., **Hadland, N.**, Sankar, R., and Nodolski, N. "Simulating the Uranian Berg Feature as a Vortex-Driven Cloud." *AGU Fall Meeting* (2018)